

Question	Answer	Mark	Mark scheme	Additional guidance
20 (a)	3.5	P1	for process to find the common difference between the first and second and the common difference between the third and second term eg $(3x + 1) - (x + 2) (= 2x - 1)$ and $(x + 2) - (x - 4) (= 6)$ or for process to write a correct equation in x, eg $(3x + 1) - (x + 2) = (x + 2) - (x - 4)$ or $2x - 1 = 6$ oe	r can be any letter apart from y
(b)	-0.5, 8	A1	for 3.5 oe eg $\frac{7}{2}$	
		P1	for process to write a correct equation in terms of the common ratio and y eg $r(y - 4) = y + 2$ or $r(y + 2) = 3y + 1$	
		P1	for process to write a correct equation in y eg $\frac{3y + 1}{y + 2} = \frac{y + 2}{y - 4}$ oe	
		P1	for process to write a correct equation without fractions eg $(3y + 1)(y - 4) = (y + 2)(y + 2)$ oe	
		P1	for process of writing a correct simplified equation eg $2y^2 - 15y - 8 (= 0)$	
		A1	cao	The quadratic does not have to equal 0, ie accept $2y^2 - 15y = 8$